Appendix A1 - South-west Corner Marine Park (Western arm) Natural Values Report

South-west Marine Parks Network

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Natural Values

Understanding of natural values

Sampling summary

Table A1.1: Summary of monitoring campaigns in the study area, showing the survey dates, sampling method, management zones sampled, and the number of samples with each data type. NPZ = National Park Zone, HPZ = Habitat Protection Zone, MUZ = Multiple Use Zone, SPZ = Special Purpose Zone.

					Habitat	Fish	
Date	**Campaign name**	**Method**	**Areas**	**Samples**	**Count**	**Count**	**Length**
Jun 2020	2020-06 south-west stereo-BRUVs	BRUV	NPZ	42	40	39	33
Oct-Nov 2020	2020-10 south-west stereo-BRUVs	BRUV	Coastal waters, NPZ, SPZ	275	273	272	244
Oct-Nov 2020	2020-10 south-west BOSS	BOSS	Coastal waters, NPZ, SPZ	262	262	-	-
Mar 2021	2021-03 West-Coast BOSS	BOSS	Coastal waters, NPZ	154	154	-	-
Mar 2023	2023-03 Augusta stereo-BOSS	BOSS	Coastal waters, NPZ, SPZ	73	70	-	-
Mar 2023	2023-03 SwC BOSS	BOSS	NPZ	88	88	-	-
Mar 2023	2023-03 SwC stereo-BRUVs	BRUV	NPZ	32	32	32	32
Oct 2024	2024-10 SWC stereo-BRUVs	BRUV	Coastal waters, NPZ, SPZ	190	190	188	178
Nov 2024	2024-11 SwC stereo-BOSS	BOSS	Coastal waters, MUZ, NPZ, SPZ	137	117	-	-
Mar-Apr 2025	2025-04 South-west Corner BOSS	BOSS	NPZ, SPZ	210	210	-	-
Apr 2025	2025-04 South-west-corner stereo-BRUVs	BRUV	NPZ, SPZ	150	148	143	137

Benthic ecosystem extent

Observations

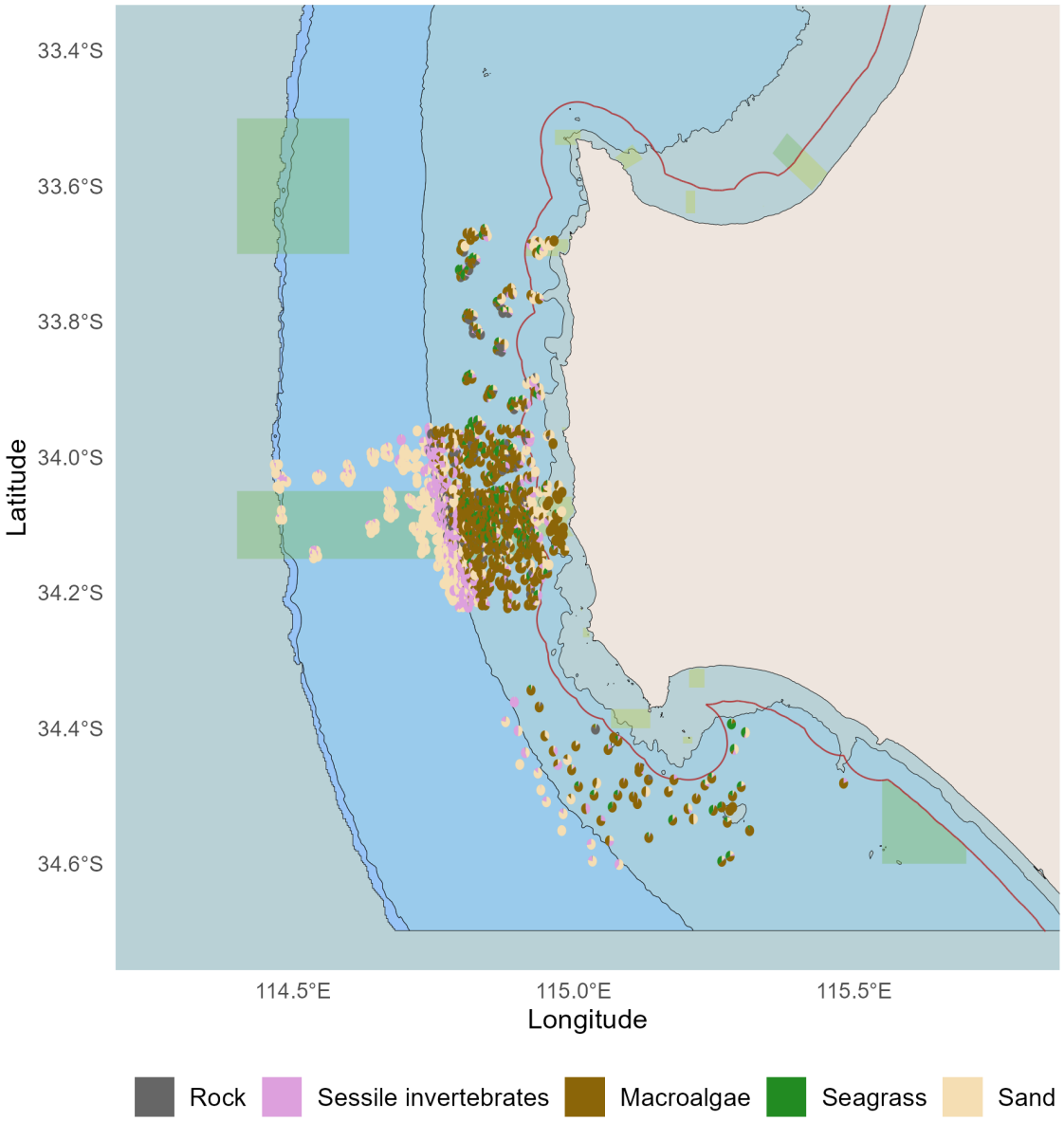


Figure A1.1: Habitat classes observed in spatially balanced stereo-BRUV and drop-camera imagery visualised as spatial pie charts. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Dominant habitat types

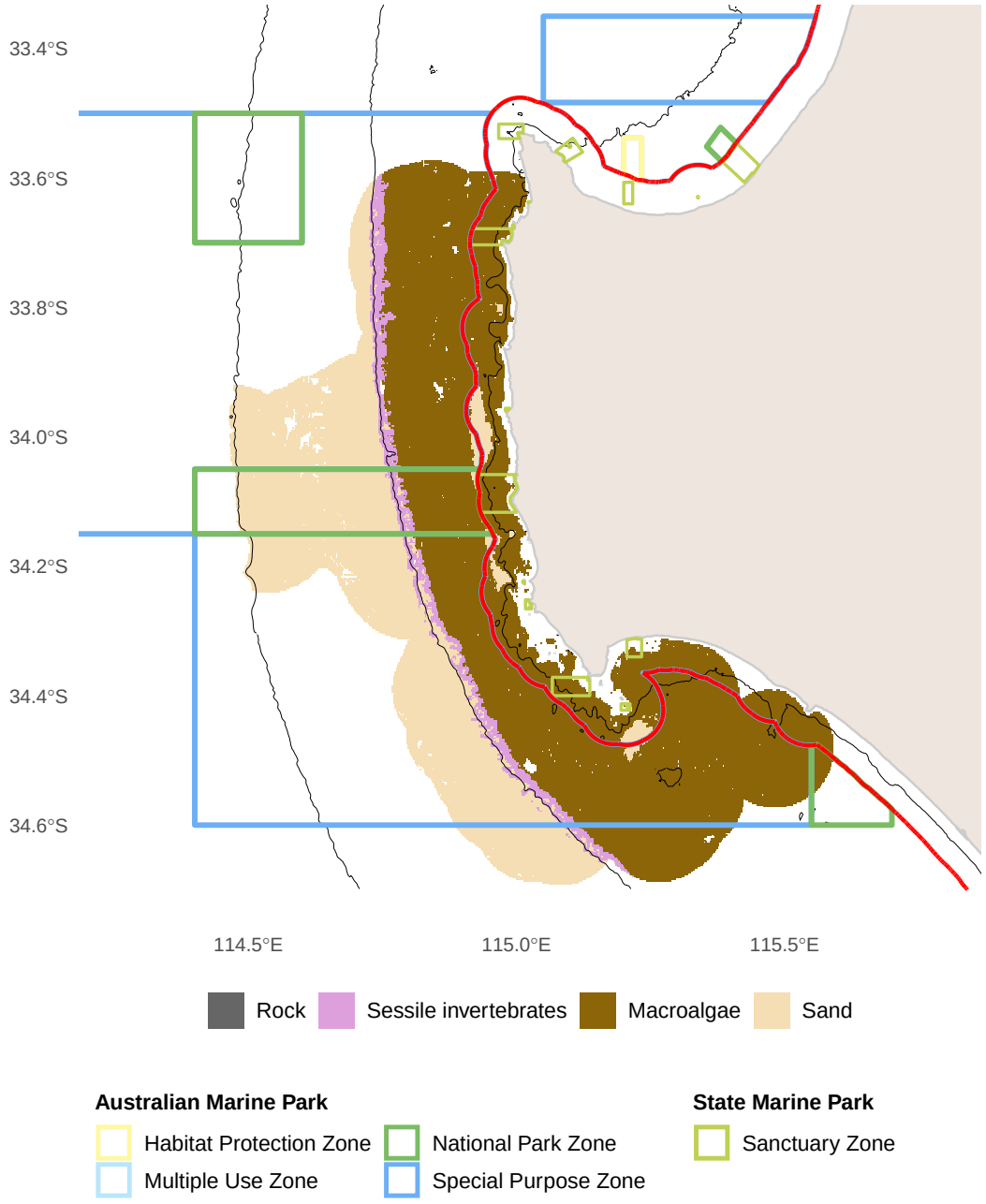


Figure A1.2: Predicted dominant habitat type from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Habitat distribution and probability

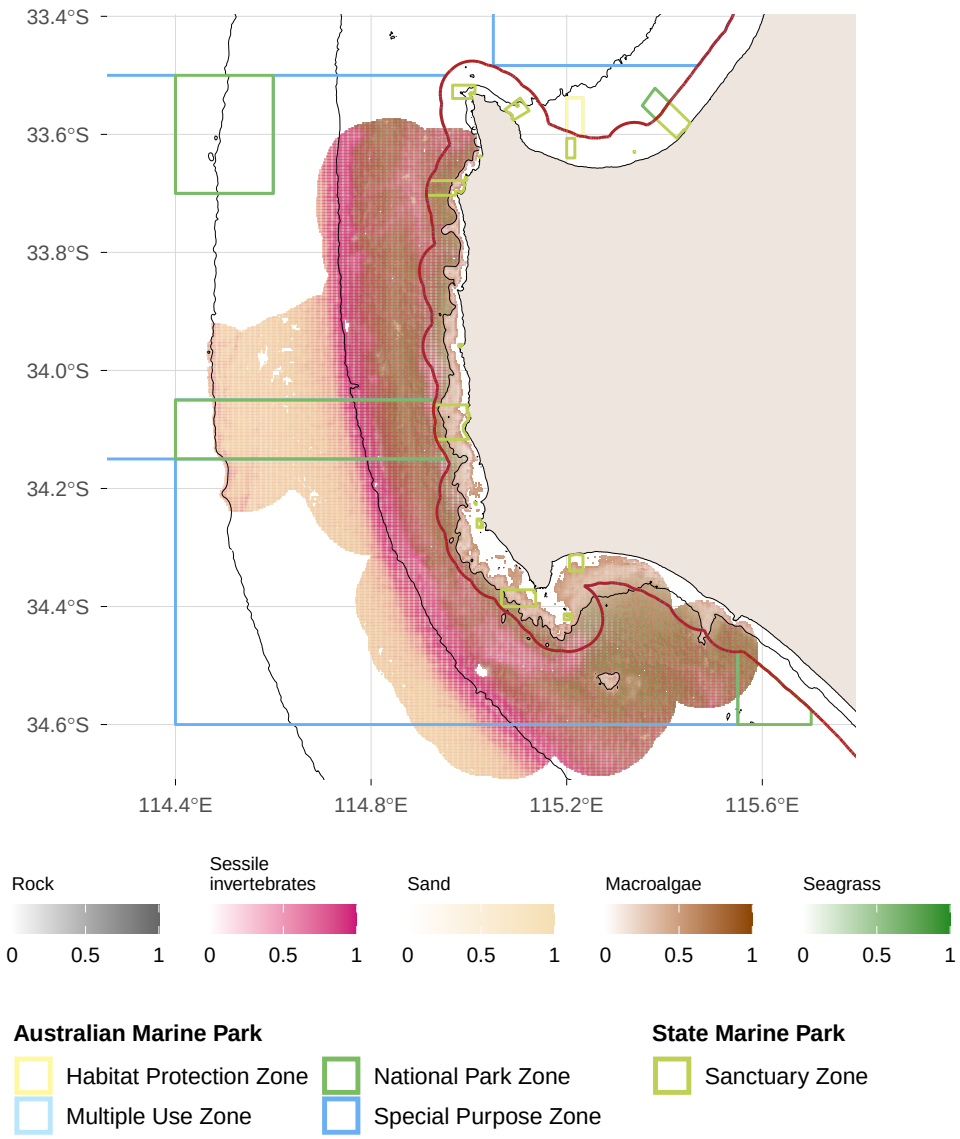


Figure A1.3: Predicted habitat distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Funcional reef distribution and probability

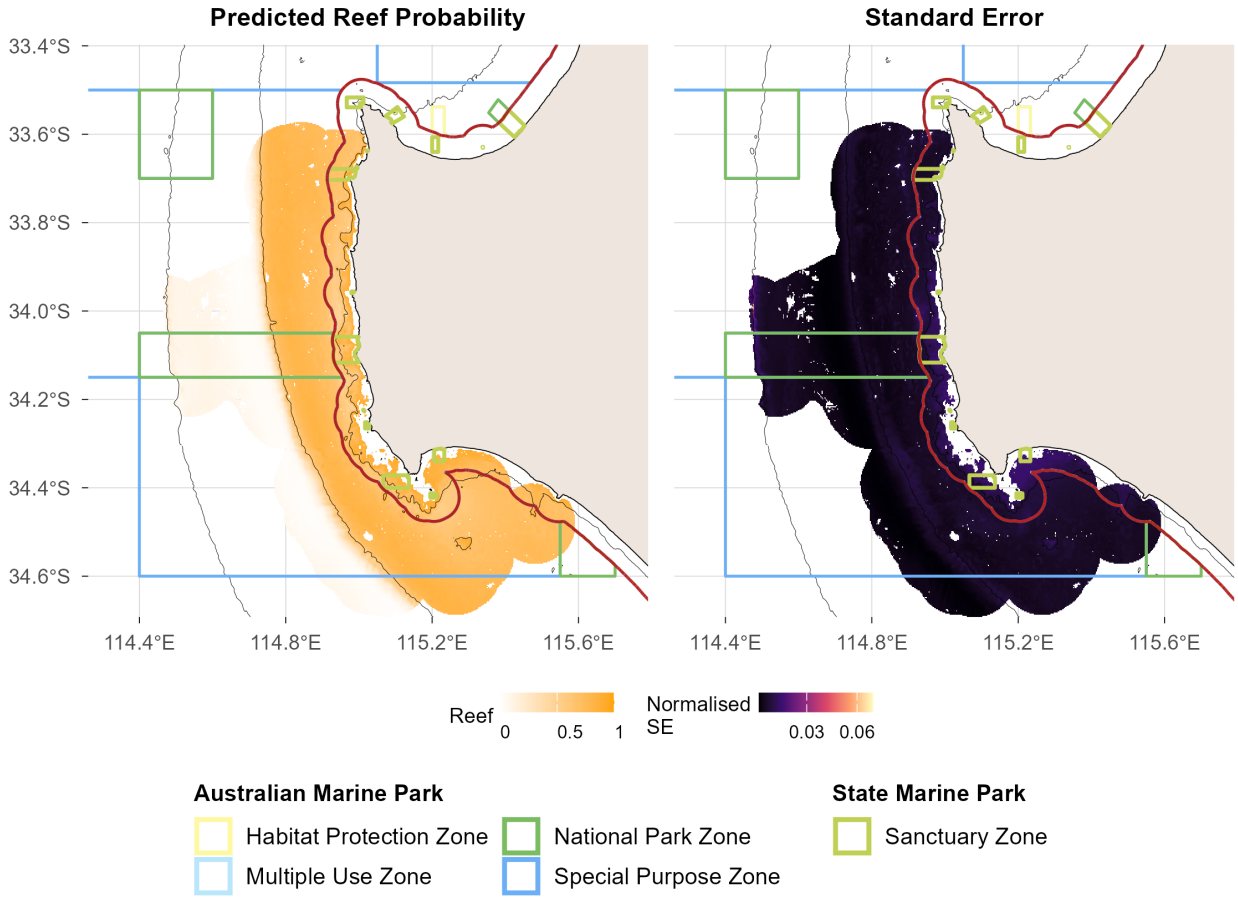


Figure A1.4: Predicted reef distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Benchmarks of benthic natural values

Sand

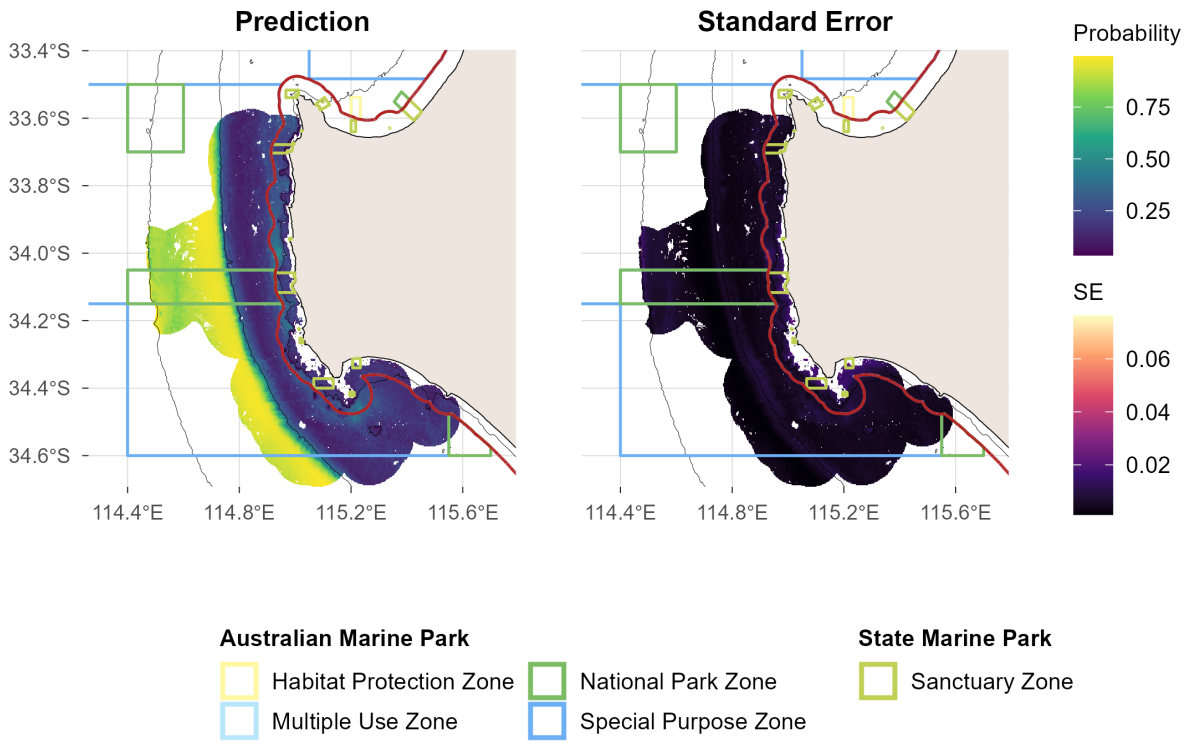


Figure A1.5.1: Sand: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Macroalgae

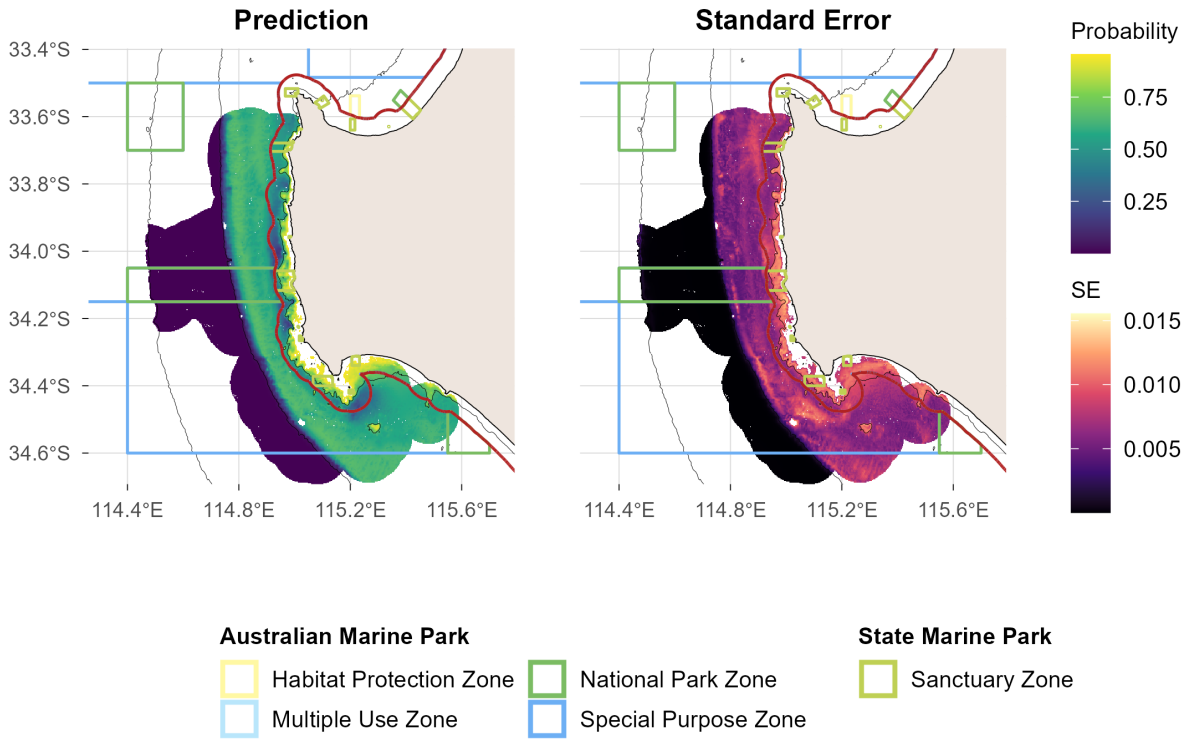


Figure A1.6.1: Macroalgae: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Seagrass

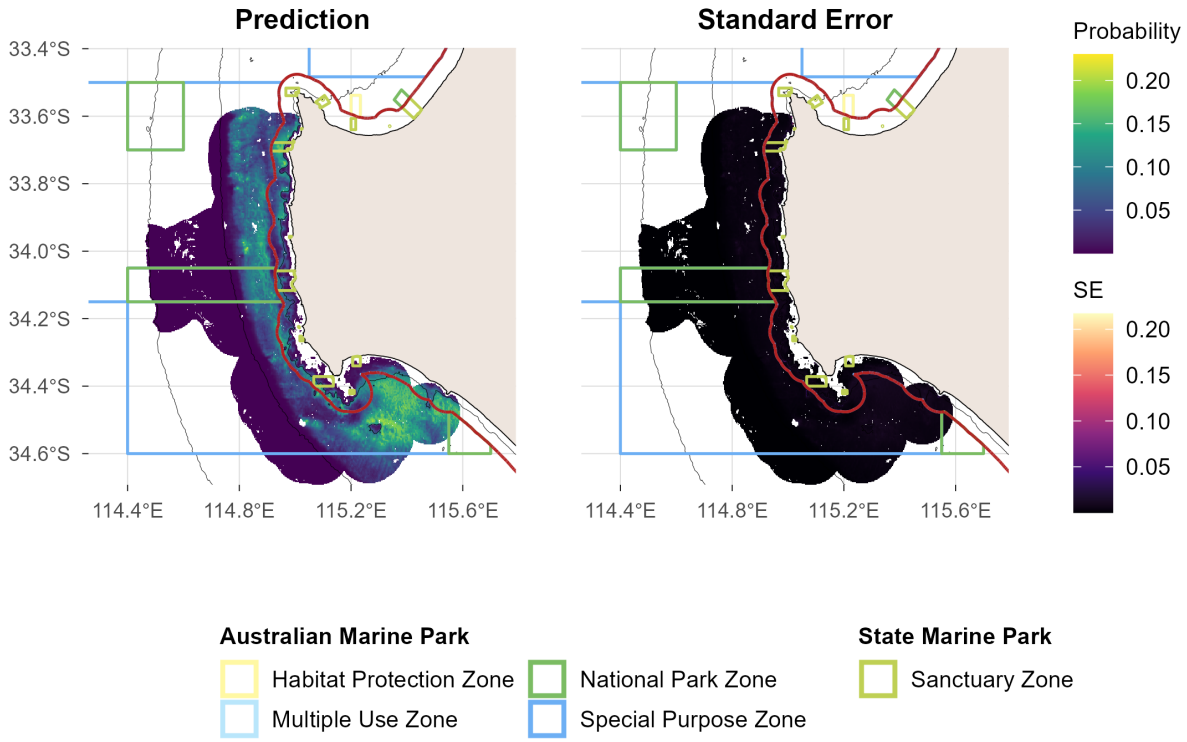


Figure A1.7.1: Seagrass: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Sessile invertebrates

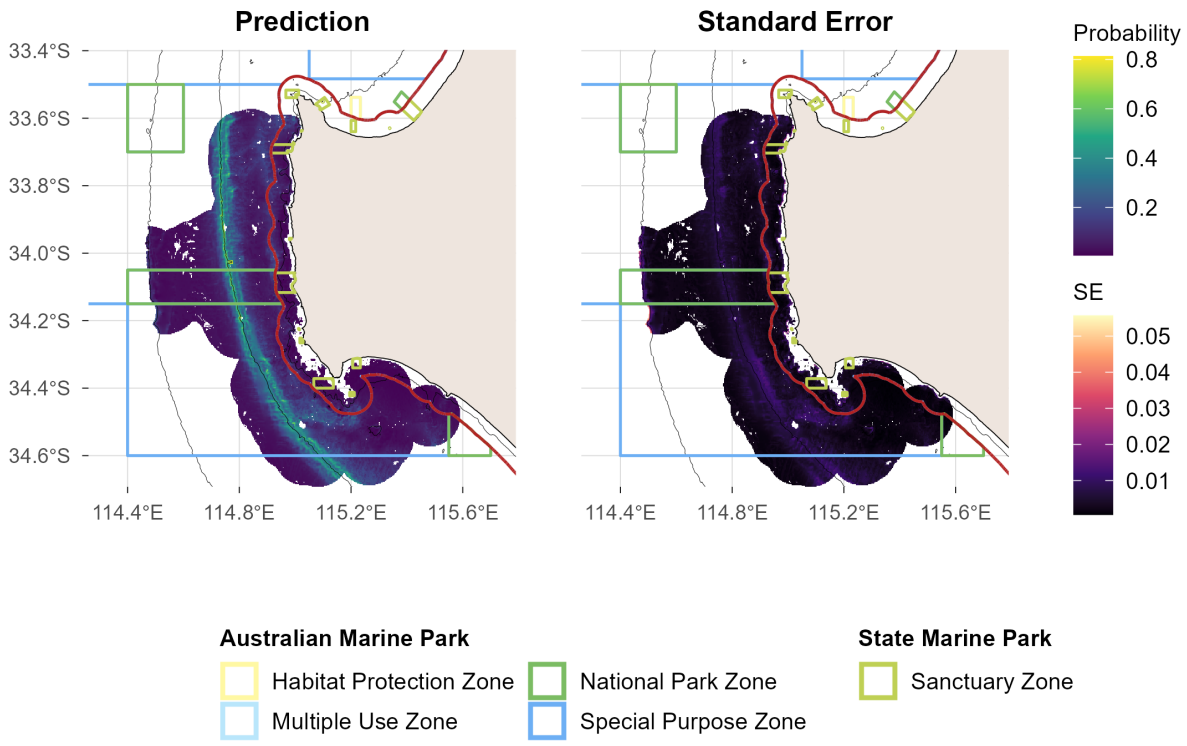


Figure A1.8.1: Sessile invertebrates: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Rock

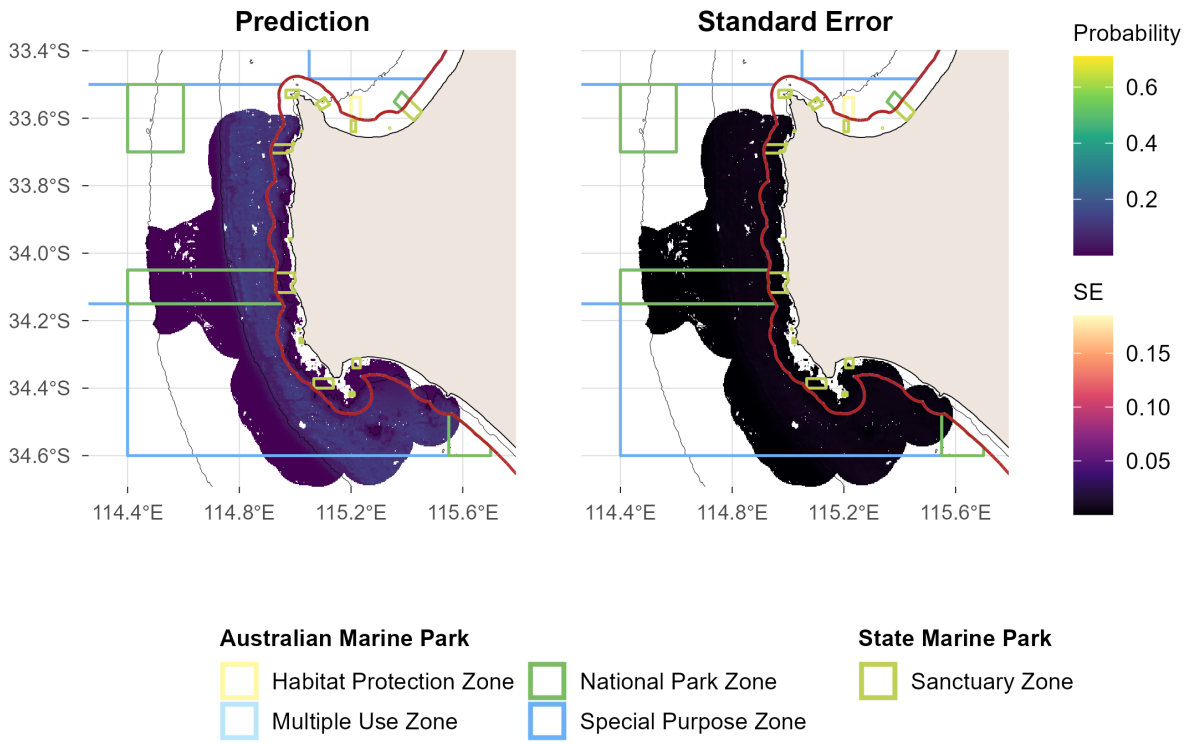


Figure A1.9.1: Rock: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Benchmarks of demersal fish natural values

Survey effort

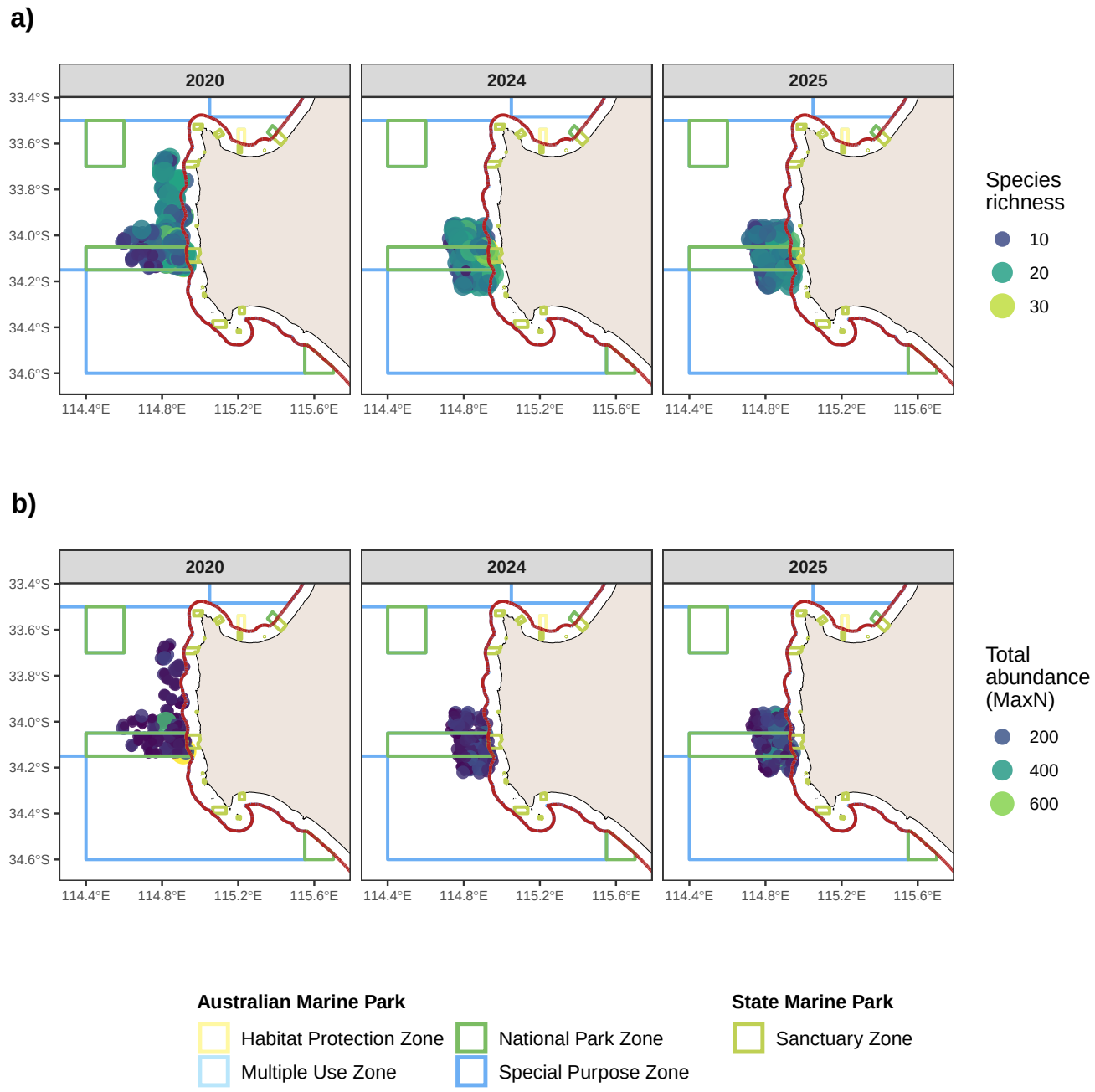


Figure A1.10: Spatial distribution of observed species richness and total abundance (MaxN) per stereo-BRUV drop, pooled across all deployments. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters.

Species richness

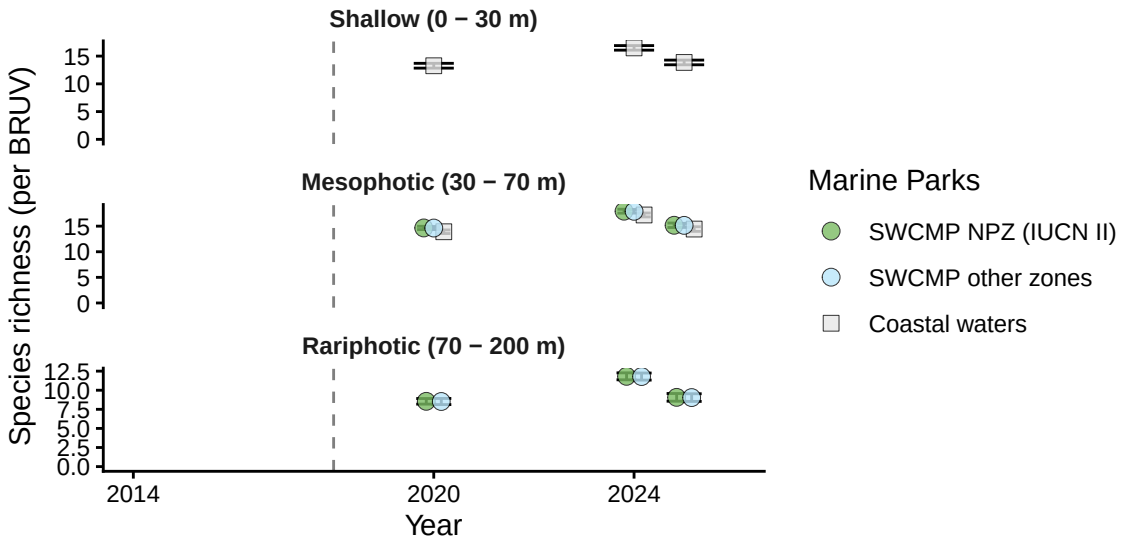


Figure A1.11.1: Demersal fish - Species richness: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

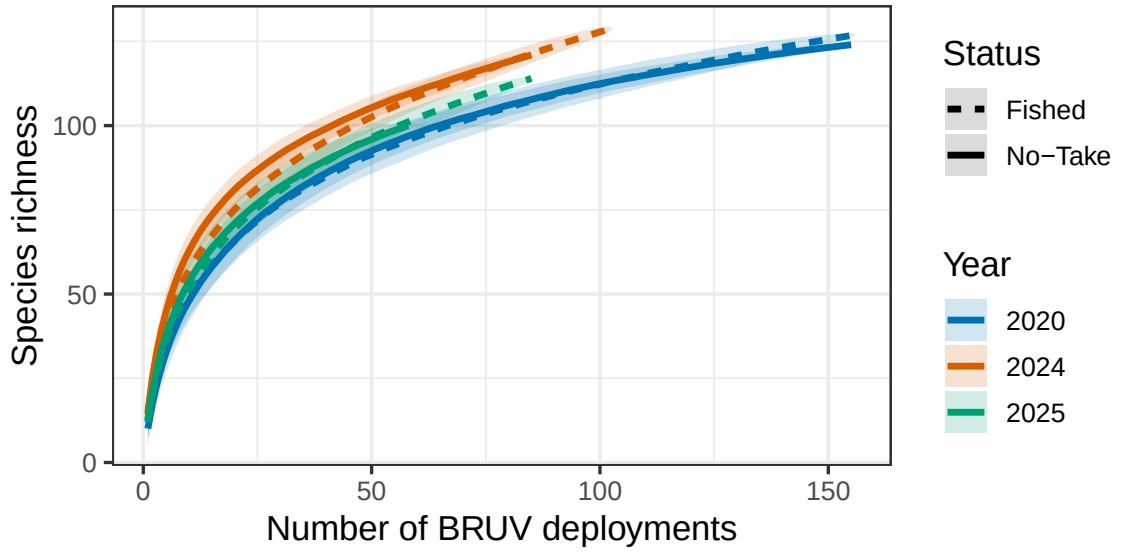


Figure A1.11.2: Demersal fish - Species richness: species accumulation curve by year from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

Species richness

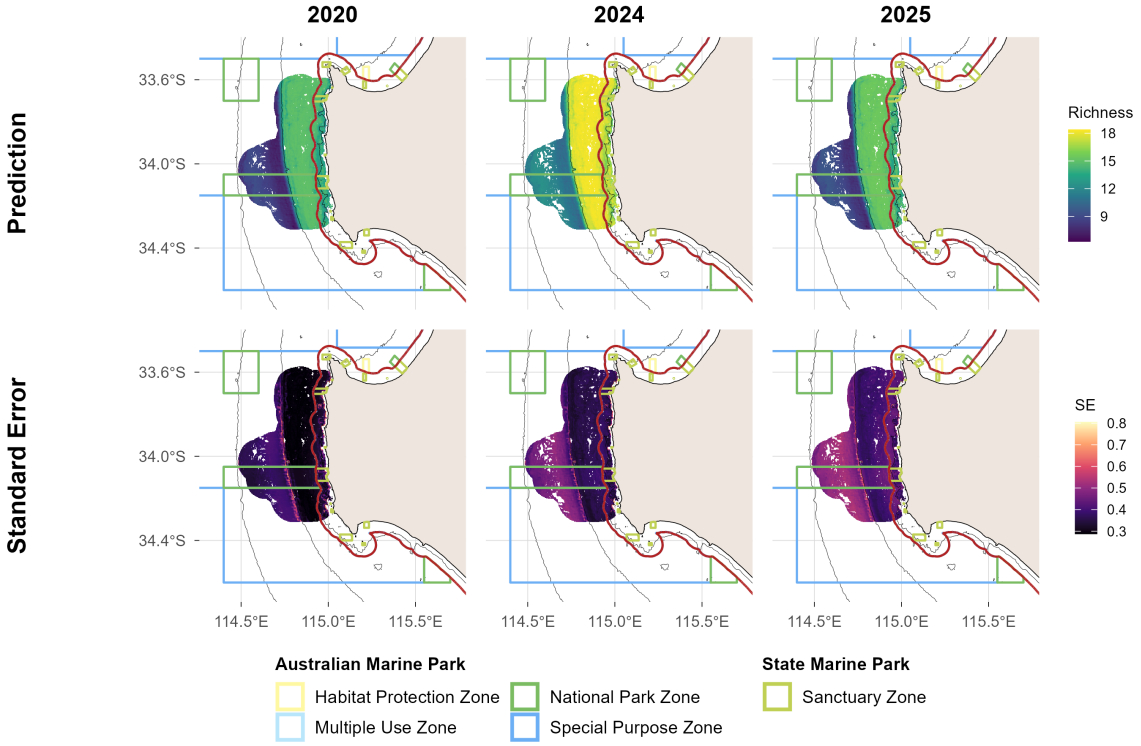


Figure A1.11.3: Demersal fish - Species richness: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Total abundance

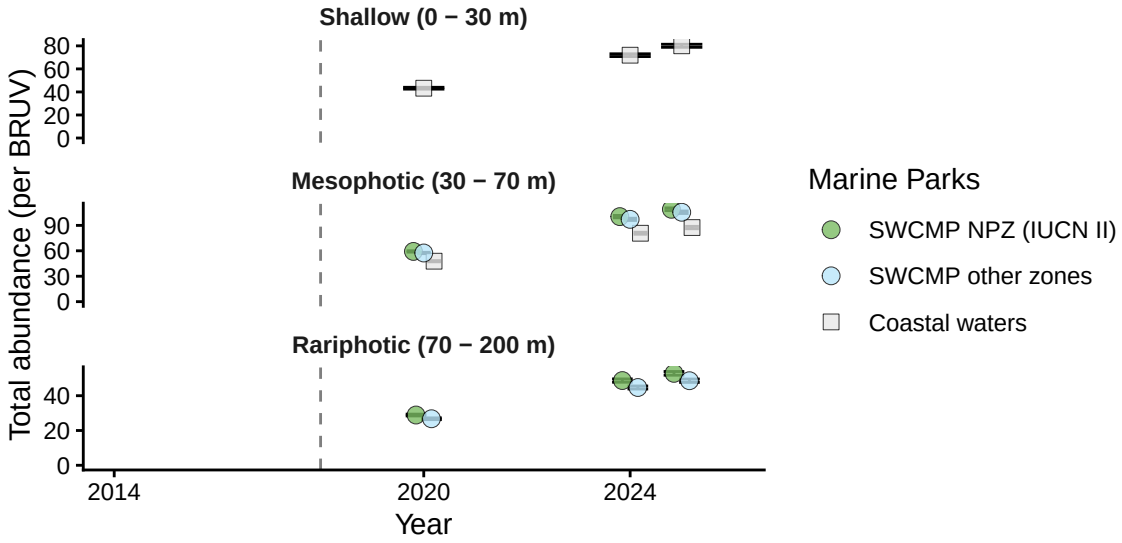


Figure A1.12.1: Demersal fish - Total abundance: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

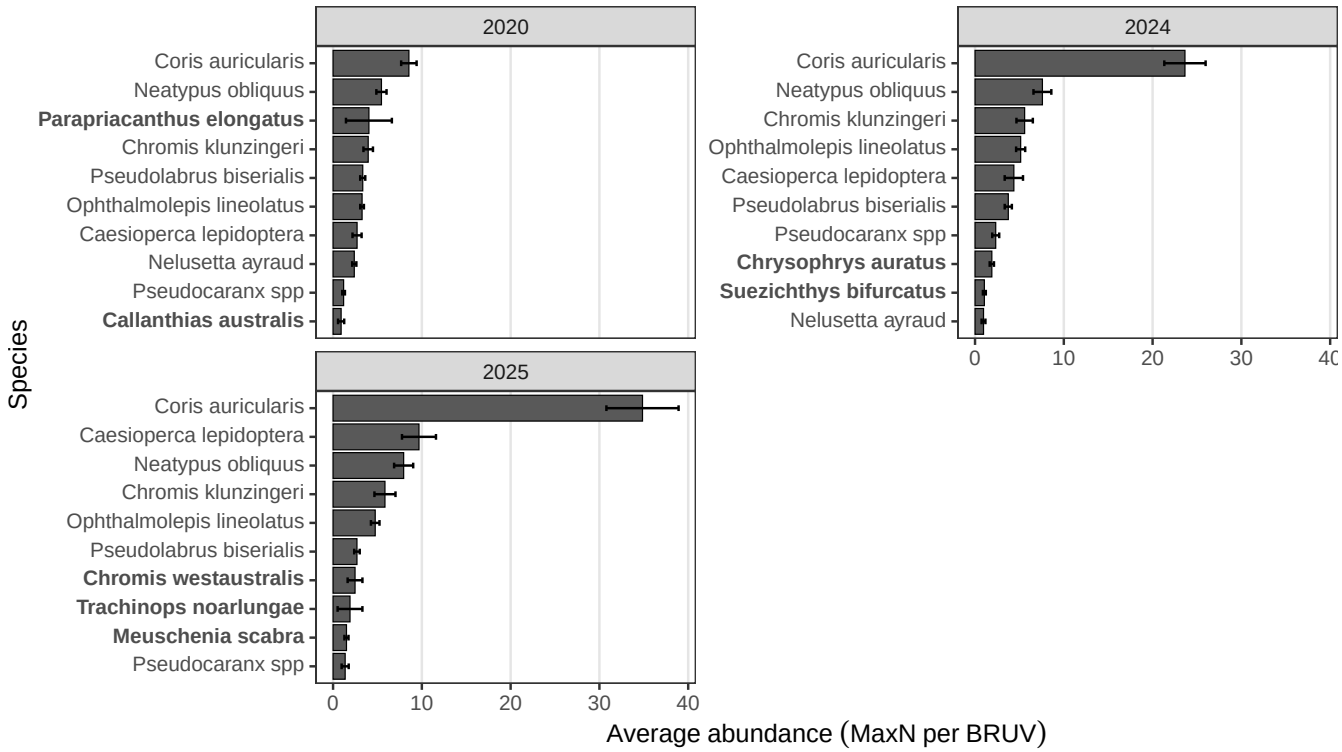


Figure A1.12.2: Demersal fish - Total abundance: top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

Total abundance

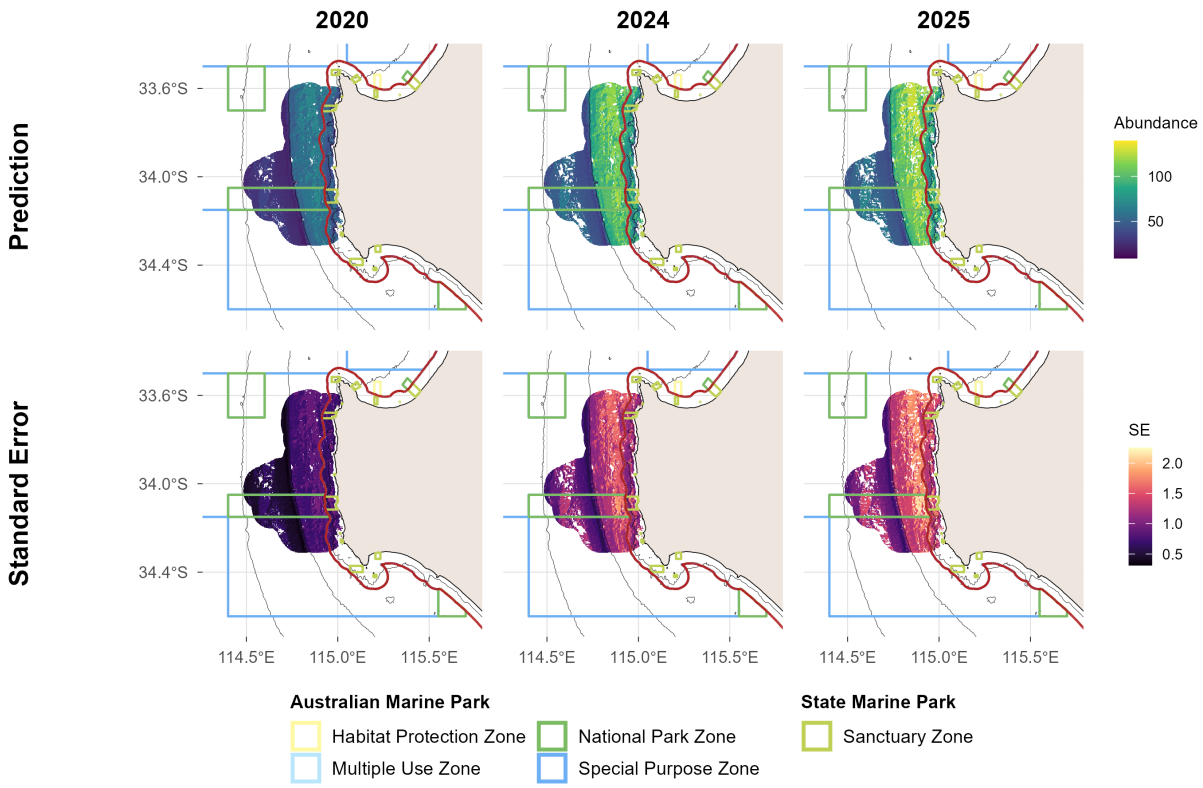


Figure A1.12.3: Demersal fish - Total abundance: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Large Reef Fish Index*

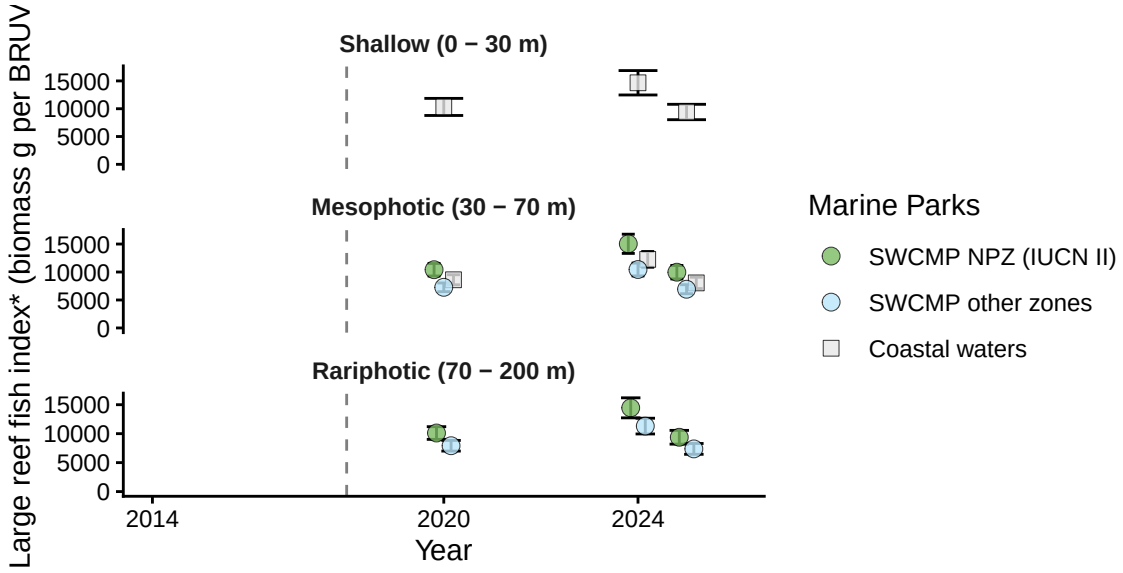


Figure A1.13.1: Demersal fish - Large Reef Fish Index*: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

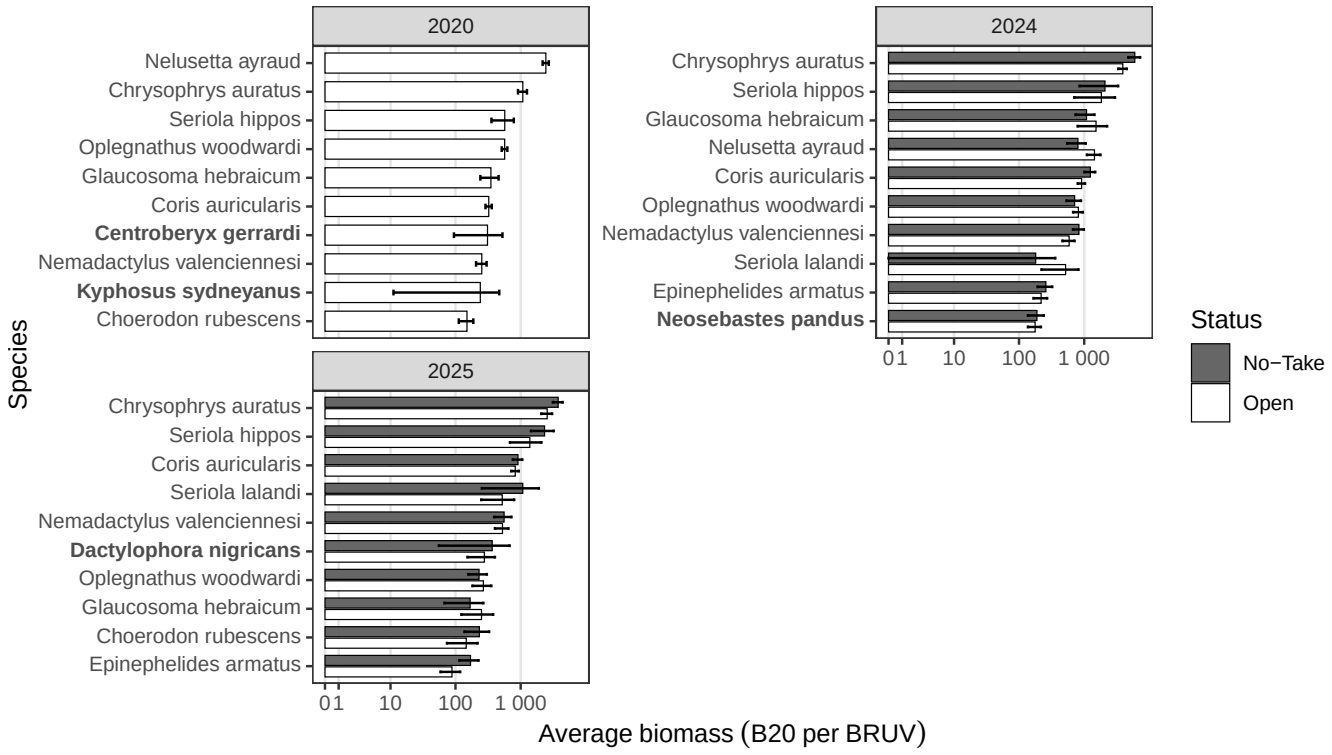


Figure A1.13.2: Demersal fish - Large Reef Fish Index*: top 10 greatest biomass species (pooled across deployments) from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

Large Reef Fish Index*

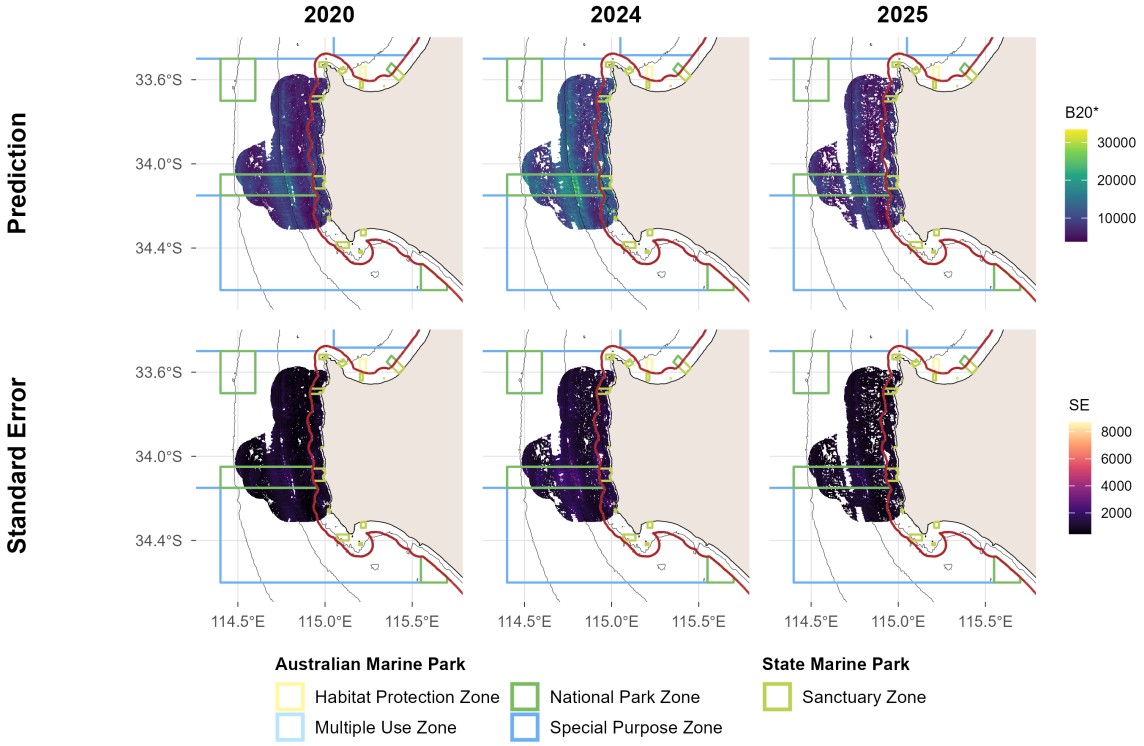


Figure A1.13.3: Demersal fish - Large Reef Fish Index*: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Reef fish thermal index

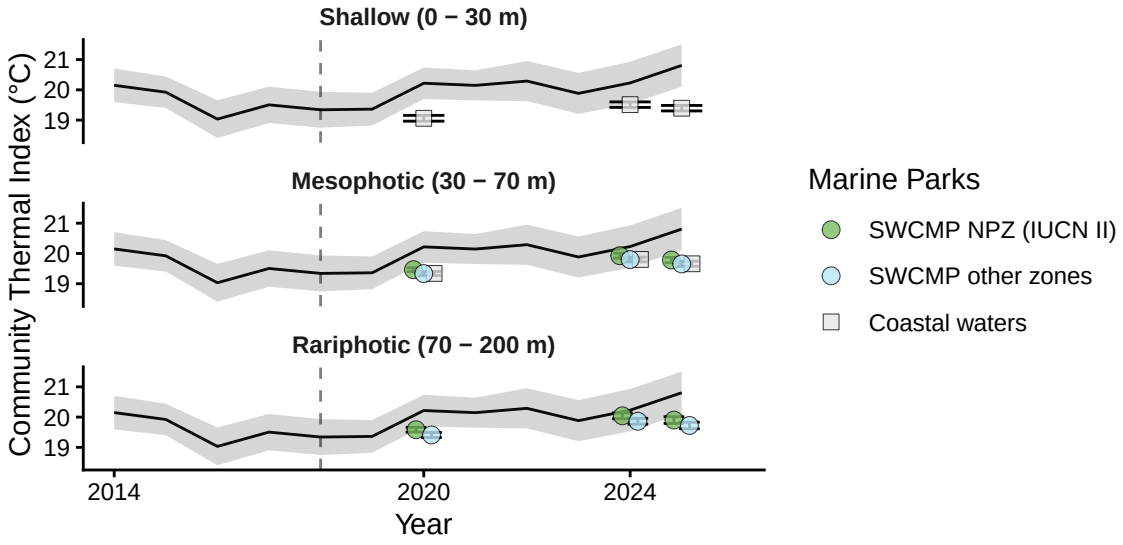


Figure A1.14.1: Demersal fish - Reef Fish Thermal Index: benchmark plots by zone and depth contour, with monthly sea surface temperature (black line, grey band = SD) overlaid, from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

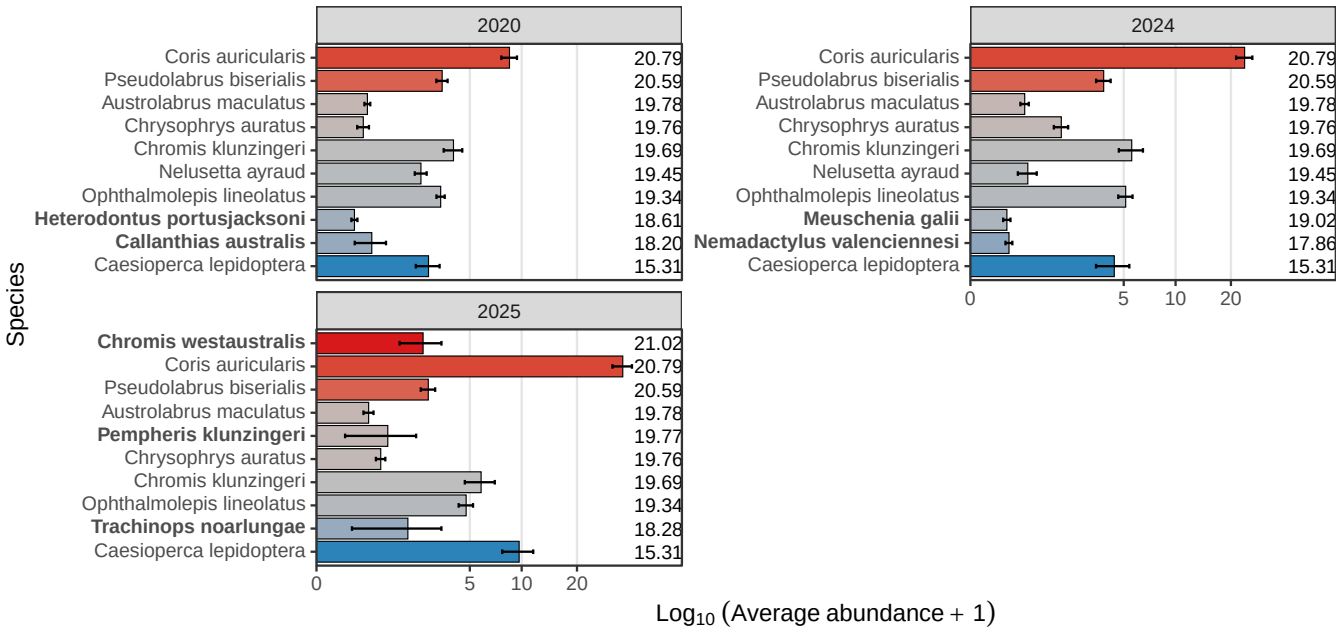


Figure A1.14.2: Demersal fish - Reef Fish Thermal Index: species thermal niche of top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

Reef fish thermal index

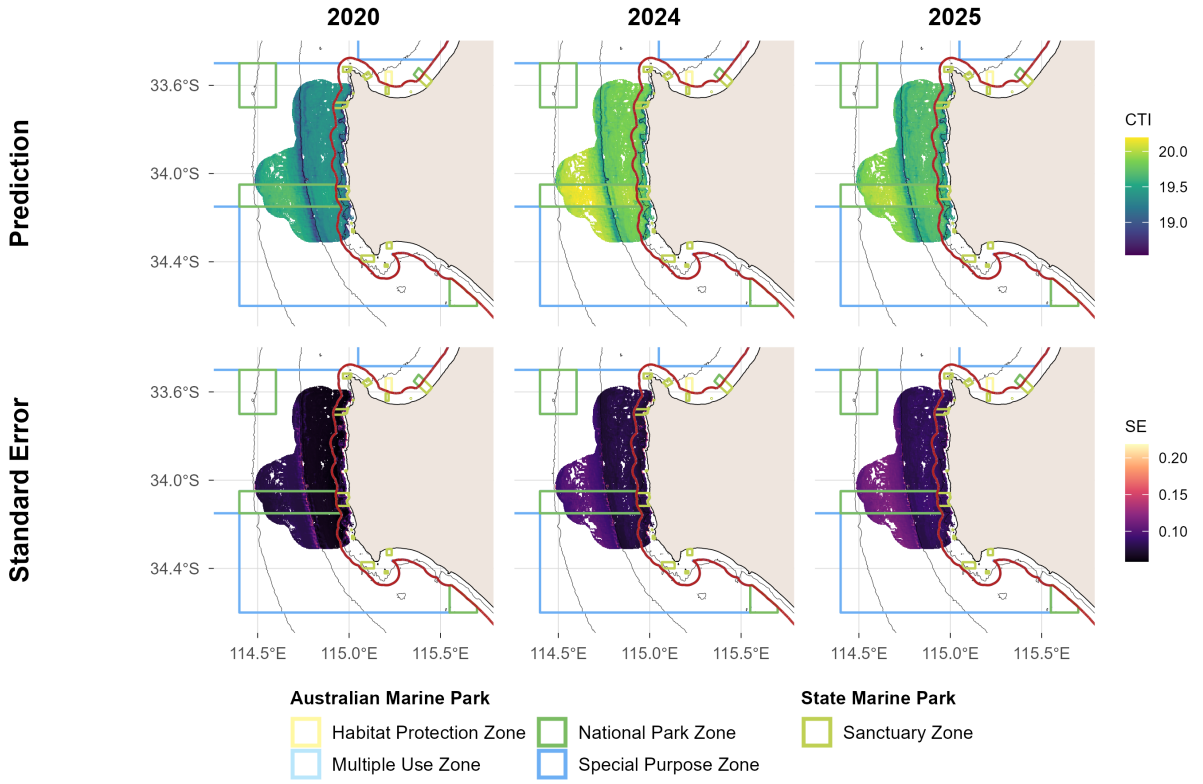


Figure A1.14.3: Demersal fish - Reef Fish Thermal Index: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Threatened species

Table A1.2: Threatened species recorded in stereo-BRUV imagery across all survey years, showing IUCN Red List category, the number of samples each species was recorded in, and the total abundance (MaxN) summed across those samples. *This species is also listed as 'Conservation dependent' on the EPBC Act List of Threatened Fauna. ^This species is also listed as 'Vulnerable' on the EPBC Act List of Threatened Fauna.

Species	Common name	Family	IUCN category	Samples	Total MaxN	Years recorded
Epinephelides armatus	Breaksea Cod	Serranidae	Near Threatened	122	162	2020, 2023, 2024, 2025
Bodianus frenchii	Foxfish	Labridae	Near Threatened	99	116	2020, 2023, 2024, 2025
Sepia apama	Giant cuttlefish	Sepiidae	Near Threatened	37	44	2024, 2025
Squalus acanthias	Whitespotted Dogfish	Squalidae	Vulnerable	18	28	2020
Achoerodus gouldii	Western Blue Groper	Labridae	Vulnerable	14	14	2020, 2023, 2024, 2025
Carcharhinus brevipinna	Spinner Shark	Carcharhinidae	Vulnerable	13	13	2020, 2023, 2024, 2025
Carcharhinus obscurus	Dusky Whaler	Carcharhinidae	Endangered	9	12	2025
Carcharhinus plumbeus	Sandbar Shark	Carcharhinidae	Endangered	7	7	2020, 2024, 2025
Galeocerdo cuvier	Tiger Shark	Carcharhinidae	Near Threatened	6	6	2024, 2025
Carcharhinus limbatus	Common Blacktip Shark	Carcharhinidae	Vulnerable	4	5	2020
*Sphyrna zygaena**	Smooth Hammerhead	Sphyrnidae	Vulnerable	4	4	2020, 2024
*Galeorhinus galeus**	School Shark	Triakidae	Critically Endangered	2	2	2020
Carcharhinus altimus	Bignose Shark	Carcharhinidae	Near Threatened	1	2	2025
Carcharias taurus	Greynurse Shark	Carchariidae	Critically Endangered	1	1	2024
Rhynchobatus australiae	Whitespotted Guitarfish	Rhinidae	Critically Endangered	1	1	2025
*Sphyrna lewini**	Scalloped Hammerhead	Sphyrnidae	Critically Endangered	1	1	2024

Threatened shark and ray species

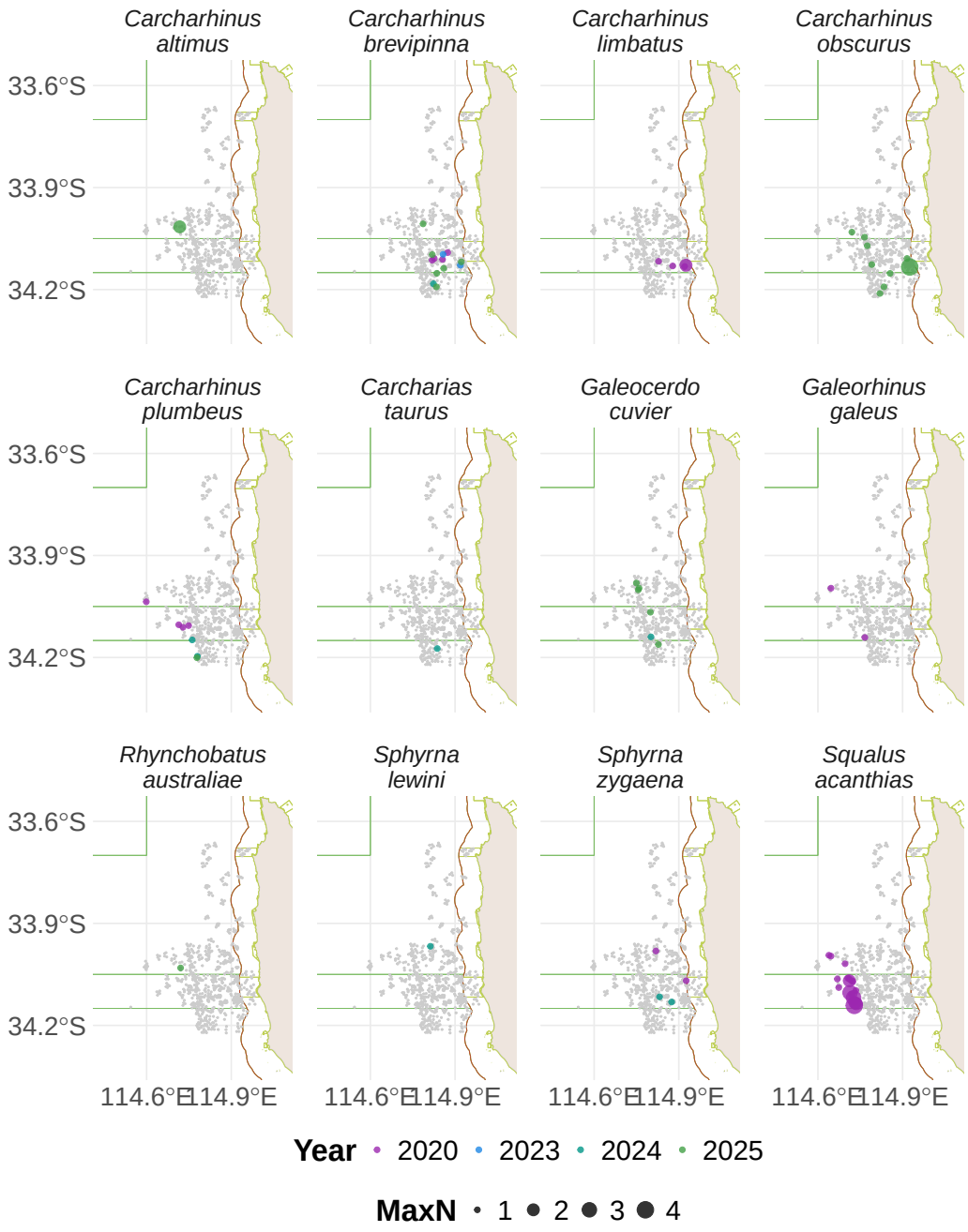


Figure A1.15.1: Occurrence of threatened, IUCN Red List category, sharks and rays in stereo-BRUV imagery. Points are scaled by abundance (MaxN) and coloured by survey year; grey points show every stereo-BRUV deployment. Commonwealth and coastal marine park boundaries are shown. The red line delimits coastal and Commonwealth waters.

Other shark and ray species

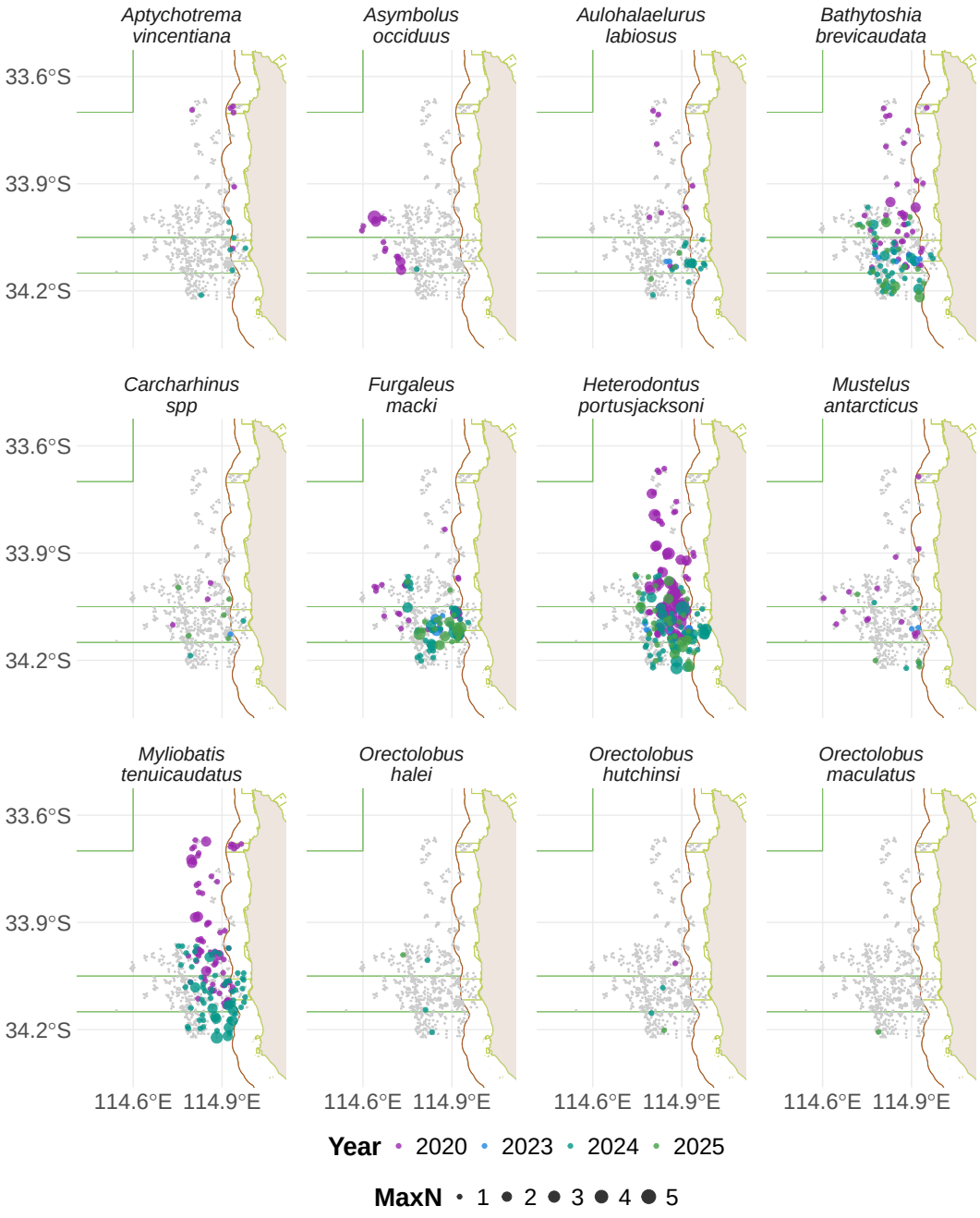


Figure A1.15.2: Occurrence of shark and ray species that are not threatened in stereo-BRUV imagery. Points are scaled by abundance (MaxN) and coloured by survey year; grey points show every stereo-BRUV deployment, so a panel with no coloured points is a genuine absence rather than an absence of sampling. Commonwealth and state marine park boundaries are shown. The red line delimits state and commonwealth waters.

Other shark and ray species

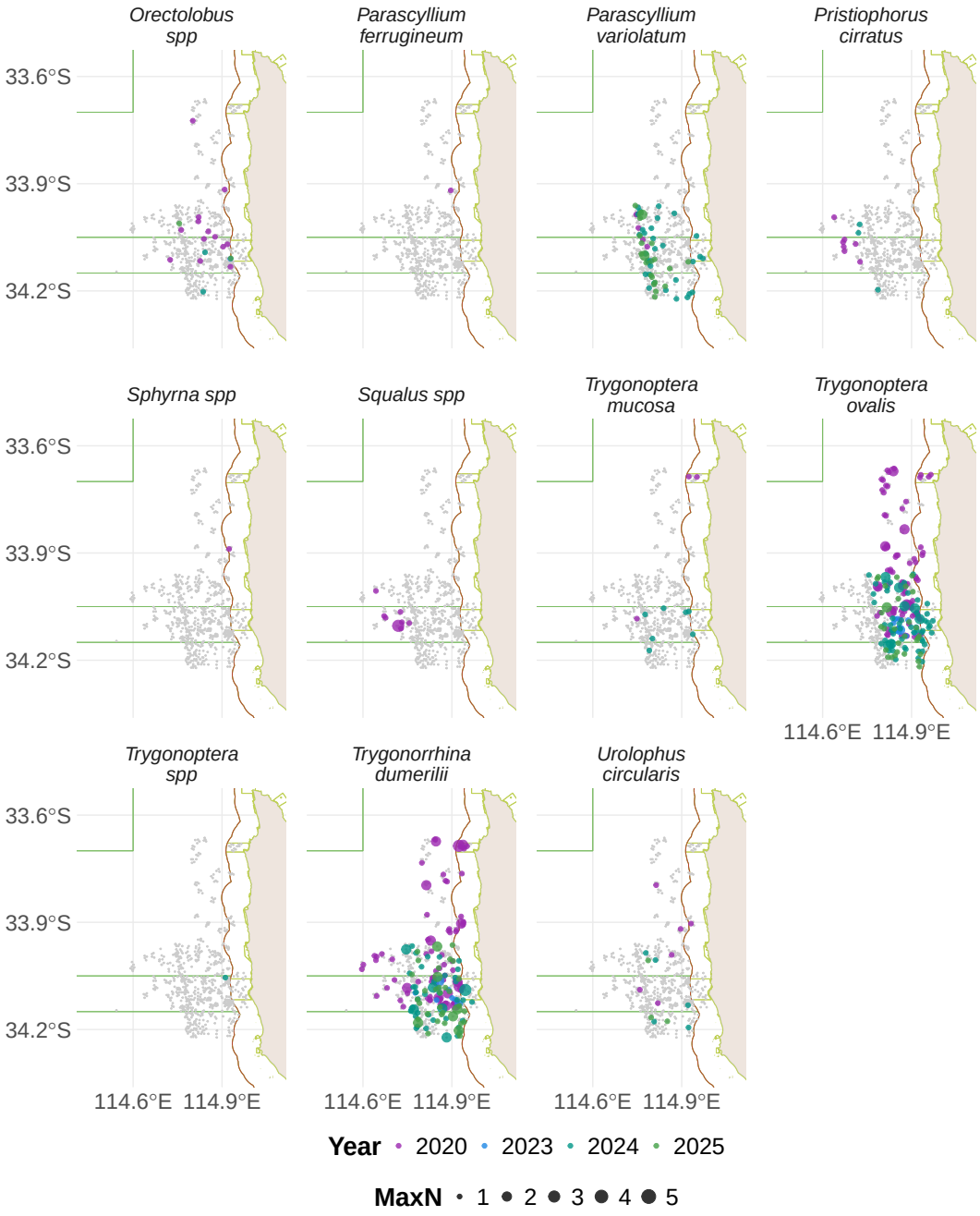


Figure A1.15.3: Occurrence of shark and ray species that are not threatened in stereo-BRUV imagery (continued). Points are scaled by abundance (MaxN) and coloured by survey year; grey points show every stereo-BRUV deployment, so a panel with no coloured points is a genuine absence rather than an absence of sampling. Commonwealth and state marine park boundaries are shown. The red line delimits state and commonwealth waters.